

Education for business analysts in Poland

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The research aim is to investigate the relationship between skills required for a business analyst's (BA) job and learning outcomes from selected Library and Information Science (LIS) degrees in Poland. Two hypotheses are stated: 1. Employers in Poland look for their future BAs among graduates with different degrees; 2. LIS-related degrees provide the core competences required for a BA job in Poland. An analysis of job offers for BAs in Poland was made, along with a comparative analysis of BA skills from the Standard Classification of Occupations, version 3 of the Guide to Business Analysis Body of Knowledge (BABOK 3), job offers and learning outcomes from Polish LIS-related degree courses. The most common requirements posted in job offers are language skills, knowledge of IT tools, communication, presentation and mediation skills, personality traits and analytical skills. Distinctive BA features such as understanding the nature of business analysis, requirements engineering, using notations and process frameworks, systems modelling and programming were found only in job offers and BABOK 3. Recommendations are made for LIS-related degree courses on how to deal with the mismatch between the skills required in the job market and those received through formal education.

Keywords: BABOK, business analysis, business analyst, job offers, learning outcomes, requirements

1. Introduction

Nowadays, according to Bonesso, Bruni and Gerli (2020, p. 21), a business analyst (BA) is considered one of the most in-demand data-driven jobs. A growing demand for professionals able to conduct business analysis is also present in Poland, though the terminology around job titles is still unclear (Jarzębowicz & Marciniak, 2017, p. 316). What does a BA do?

In general, a BA is responsible for eliciting and analysing requirements (e.g. in the form of user stories in Gherkin) and facilitating communication between a client and a team. In other words, a BA is someone who enables change in an organisation, because they identify business needs and recommend solutions that create value (Jarzębowicz & Połocka, 2017, p. 1189). A BA performs these tasks in a specific working framework (e.g. Scrum, Kanban, Waterfall) used in a project (Chernysheva & Shepelenko, 2018, p. 92). There is also a Quality Analyst (QA) who deals with testing requirements, products and processes. Sometimes, the BA and QA are linked together in the form of a Business Quality Analyst (BQA) (e.g. Ahmed et al., 2019, pp. 301–302). All BAs, QAs and BQAs are classified alongside other service-oriented occupations, such as information brokers, librarians and many other information professionals and knowledge workers, because the core of these jobs is mediation in order to provide

the best possible content or product for the right audience (Wojciechowski, 2014, pp. 59–60).

Based on the Standard Classification of Occupations, a BA in Poland is generally considered a professional with a degree. The code of a BA – 242112 – shows this, since the number two at the beginning of the code is reserved in this classification for a specialist with at least level five of the International Standard Classification of Education (Ministry of Labour and Social Policy, 2018). However, there are no regulations with regard to any specific degree required to become a BA in Poland. In cases like this, while selecting a degree in preparation for a BA position, prospective students might use several sources to help them make an informed choice. One can use, for example popular online guides, such as www.OtoUczelnie.pl or www.PolMaurze.pl, and queries in global search engines.

As of June 2020, such sources suggest a few first and second cycle degree courses and many more options for postgraduate education. For the first cycle, one can find Business Analytics (offered by the Faculty of Organisation and Management, Silesian University of Technology) and Information Technology with a specialisation in Business Analytics and Implementation of IT Systems (WSB University, Chorzów). As for the second cycle, suggestions include Information Systems Engineering (Department of Software Engineering, Faculty of Electronics, Telecommunications and Informatics, Gdańsk University of Technology), Finance and Accounting accredited by the Association of Chartered Certified Accountants (ACCA) (SGH Warsaw School of Economics) and Management with a specialisation in Business Analytics (Faculty of Management, University of Warsaw). Finally, postgraduate education in Business Analytics courses is offered by (1) the School of Banking and Management in Krakow, (2) the Poznań University of Economics and Business, (3) the Institute of Economics and Finance in the Faculty of Social Sciences, Catholic University of Lublin, and (4) the Institute of Computer Science in the Faculty of Electronics and Information Technology, Warsaw University of Technology. There are also postgraduate education courses labelled (5) Business Analytics for Managers (University of Information Technology and Management in Rzeszow), (6) Analysis of Business Processes (Tischner European University) and (7) Requirements Management and Business Analysis in Projects (Kozminski University). Moreover, (8) Kozminski University offers courses in cooperation with the International Institute of Business Analysis (IIBA) preparing for the Certification of Competency in Business Analysis.

As can be seen from this overview, although suggested degree courses are often labelled Business Analytics, they are offered by various faculties and departments, usually specialising in either Informatics, Economics or Management. Taking into account this variety of disciplines, a question arises: apart from the technical relevance of names of degree courses, are there any other studies useful for prospective BAs? For example, degrees in the fields of data analysis and data science, which are now developing in Poland, could be considered. The Sector Council on Skills, a part of the Integrated Qualifications System developed by the Polish Educational Research Institute and the Ministry of National Education, mentions data analysis degrees

such as Engineering and Data Analysis (AGH), Mathematical Modelling and Data Analysis (University of Gdańsk), and Data Analysis – Big Data (SGH Warsaw School of Economics) (Polskie Towarzystwo Informatyczne, 2019).

What is more, in Poland, Library and Information Science (LIS)-related degree courses are evolving. And during history of LIS degrees in Poland, they have already produced several information professionals, including librarians, information brokers, information architects and user experience (UX) designers. In the recent years, significant changes have occurred in LIS education in Poland. In 2013, there was the deregulation of librarian as a profession. Then, in 2018, a regulation on the division of science, scientific and artistic disciplines was adopted as a part of the new Law on Higher Education and Science. The regulation deletes Library and Information Science (LIS) as a discipline and introduces Social Communication and Media instead.

These changes forced the present Polish LIS faculties to seek new paths in research and education. Currently, a list of Polish LIS-related faculties and degrees includes:

1. Archival Studies, Document Management and Information Brokering (first and second cycle) – Pedagogical University of Krakow;
2. Information Architecture (first cycle) – Pedagogical University of Krakow;
3. Information in Digital Environment (first cycle) – Department of Information, Library and Book Science, Faculty of Philology, University of Lodz;
4. Information Management (first and second cycle) – Institute of Information Studies, Faculty of Management and Social Communication, Jagiellonian University (former LIS taught at the Institute of Information and Library Science);
5. Information Management and Digital Publishing (first and second cycle) – Pedagogical University of Krakow;
6. Information Science with Business English (second cycle) – Department of Information, Library and Book Science, Faculty of Philology, University of Lodz;
7. LIS (first and second cycle) – Faculty of Humanities, University of Silesia; Faculty of Journalism, Information and Book Studies, University of Warsaw; Institute of Library and Information Science, Faculty of Philology, University of Wrocław;
8. LIS (first cycle) – Adam Mickiewicz University, Poznań?

One can obviously become a BA after graduating with LIS-related studies, because students come from different educational backgrounds and can also acquire skills outside the formal system of education. What is more pressing is responding to the following two questions: (1) are LIS-related degrees suitable for prospective BAs? (2) how can LIS-related studies be transformed into a better degree course for prospective BAs?

Job requirements from BA job offers and their ties with university education happen to be the subject of scientific projects (e.g. Verma et al., 2019). Knowing the demands of employers is necessary when creating a graduate profile (e.g. Wichrowska et

al., 2018, p. 66). A graduate profile is usually understood as a set of competencies acquired by the graduate in terms of knowledge and skills (e.g. Martinez-Vazquez, 2019, pp. 119–121; Opatha, 2019, p. 2367; Santoso & Hadi Putra, 2017, p. 146). It is undoubtedly impossible to develop a good degree course without envisioning the graduate profile (e.g. Brouwer et al., 2020, pp. 225–226; Opatha, 2019, p. 2367), therefore an understanding of employers' expectations regarding skills required of ideal job candidates should be taken into consideration when redesigning any course.

Degree course changes may become inevitable for many universities in order to attract prospective students, due to the fast-changing job market and employers' demands. As stated in Santoso and Hadi Putra (2017, p. 145), there is a growing mismatch between university education and market demands, especially in IT. Moreover, job classifications and certification schemes often do not follow job market changes fast enough (Santoso & Hadi Putra, 2017, p. 147). This is a problem faced in Poland, where the Ministry of Family, Labour and Social Policy is responsible for the Standard Classification of Occupations. Although the Classification is updated every two years, it still seems too slow for the Polish job market.

According to Rouvrais and Gaultier Le Bris (2018), there are two interesting solutions for more job market-oriented and flexible courses: (1) liberal arts and (2) breadth courses. Liberal arts are successfully used in the United States, although they cannot be implemented in European Union universities which operate within fixed outcome-based requirements (Rouvrais & Gaultier Le Bris, 2018, p. 327). On the other hand, breadth courses easily fit the EU higher education system. They are selected optionally by students and offer up-to-date skills and knowledge, yet are supposed to be accessible for non-specialists. One example of such a course is Business Intelligence, Deep Learning and Leadership (Rouvrais & Gaultier Le Bris, 2018, pp. 328–329).

Without a doubt, creating degree courses is a time-consuming process. In Poland, it involves preparing learning outcomes for a degree that needs to be accepted by the faculty council and the Accreditation Commission (e.g. Rocki, 2018, pp. 15–16). A degree course should also attract prospective students, as funding is connected with the number of undergraduates.

Given the changing job market and financial dependence, it is understood that faculties aim for degree course changes. Yet, each course change should be well thought out and set in the framework of the higher education system. Under no circumstances should degree course changes be undertaken in a moment of panic about losing funding, without a clear goal in mind. Unfortunately, when this happens, an unproductive and unethical situation arises in which students are given a false assurance of educational quality.

Having said that, providing a BA profile for LIS graduates seems to be an interesting choice for both LIS faculties and students. However, to do this the right way, an assessment needs to be done first of the relationship between requirements for prospective BAs and learning outcomes in selected LIS-related degrees. Comparing

those sets of skills allows us to find out how well-selected degree courses fit requirements and tasks performed by BAs. It could lead to some thought-through changes to selected Polish LIS-related degree courses, such as Information Management. This study intends to do this: to create a BA profile based on known BA standards and actual employers' requirements for prospective BAs in Poland, and then to compare such a BA profile with learning outcomes in selected LIS degrees; this will facilitate reasonable potential changes in existing Polish degree courses such as Information Management, as well as the creation of new courses, bearing in mind the best interests of students.

2. Research aim, goals and hypotheses

The research aim is to investigate the relationship between skills required for a BA's job and learning outcomes in selected LIS-related degrees in Poland.

There are four research goals for this study:

- G1. To analyse requirements specified in job offers posted on LinkedIn for early-stage BAs in Poland.
- G2. To identify business analyst skills covered in up-to-date standards: BABOK 3 (IIBA, 2015) and the Polish Standard Classification of Occupations (Ministry of Family, Labour and Social Policy in Poland, 2020).
- G3. To create a codebook for skills required for a business analyst's job, based on skills identified in analysed job offers for business analysts in Poland and two standards – BABOK 3 and the Standard Classification of Occupations.
- G4. To find out which of the skills required for a BA are taught on LIS-related Information Management degrees in Poland, based on the created codebook and analysis of current learning outcomes from the degrees.

Based on the initial knowledge of BA required skills and LIS education in Poland, two working hypotheses were formulated:

H1. Employers in Poland look for their future BAs among graduates with different degrees.

No specific degree is legally required for someone to work as a BA in Poland. It is assumed that employers are open to candidates with different backgrounds, not necessarily holding specific degrees (e.g. in Economics) as long as they possess skills required for the job.

H2. LIS-related degrees provide the core competences required for a BA job in Poland.

It is assumed that BAs and information professionals with LIS degrees have to succeed in mediating between different stakeholders, for example product owners, programmers and users. Both also evaluate and develop information products such as IT systems. These common main tasks might lead to common main skills as well. Still, specific requirements need to be investigated for a BA profile and an LIS graduate profile.

Expected products of this research include (1) a comparative analysis of BA skills required by employers in Poland which lead to the list of appropriate/desired skills for BA, (2) the characteristics of prospective employers, which could be considered to facilitate cooperation with higher education institutions, and (3) basic recommendations on how to transform selected Polish degrees to better fit market requirements for prospective BAs.

3. Method

Two procedures were used: (1) an analysis of job offers, and (2) a comparative analysis of BA skills from the Polish Standard Classification of Occupations, BABOK 3, job offers and learning outcomes for a Polish LIS-related degree course.

To examine job offers, an open-coded thematic and content analysis (e.g. Braun et al., 2019) was used. All 144 job offers were collected from LinkedIn between 4 December 2019 and 5 January 2020. The search term was 'business analyst'. The experience filter was set to beginner or internship. The location was fixed as Poland. Duplicates were eliminated, leaving the final number of 108 unique job offers.

Details describing each offer were gathered and stored in a spreadsheet file to create a data model. The main sheet contained 21 columns: offer ID, sponsored add (true/false), offer type, job name, company name, company size, location, link, company description, job description, responsibilities, requirements (must haves), additional requirements (can haves), benefits, other information, salary, experience level, type of employment, industry, LinkedIn function and source. All job requirements were extracted, transferred to a new sheet and divided into codeable features. This sheet had four columns: offer ID, feature, type (must have, or additional) and code. In spite of the open-coded approach, the data accuracy filter was set based on the list of codes from a codebook. The codebook was kept in a separate spreadsheet with all codes and their descriptions. The final codebook had 196 codes. The prepared files were then imported into Tableau for further analysis and visualisation of data, as well as to create code groups. More than one code was assigned to a job offer. Some data was missing with reference to the size of the company or the place of the employment.

The second procedure, the comparative analysis, explored skills from the following:

- (1) a BA job description from the Polish Standard Classification of Occupations (classification code 242112, Polish job name: *analityk biznesowy*) prepared by the Polish Ministry of Family, Labour and Social Policy (former Ministry of Labour and Social Policy),
- (2) version 3 of the Guide to the Business Analysis Body of Knowledge (BABOK 3),
- (3) collected job offers, and
- (4) learning outcomes for a LIS-related Information Management degree course

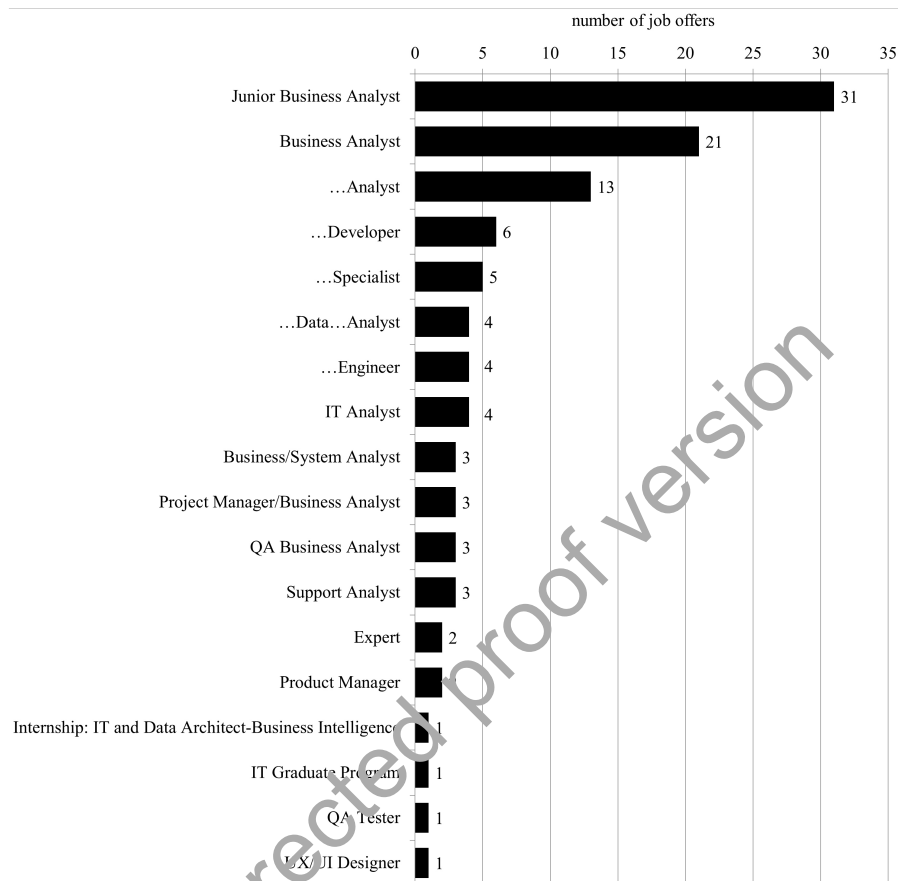


Fig. 1. Job titles in decreasing order of BA job offers. The three most common categories are Junior Business Analyst, Business Analyst and Analyst, with $n = 108$.

from one of the top Polish universities, both for a first and second cycle. The first cycle is the first level of university studies, lasts for three or four years and leads to the professional title of *licencjat* or *inżynier*. It is a Polish equivalent of the bachelor's degree. The second cycle is the second level of university studies, lasts for one and a half or two years and leads to the professional title of Master.

4. Results

This section is divided into two sub-sections, first explaining the analysis made of the job offers, followed by the results of a comparative analysis.

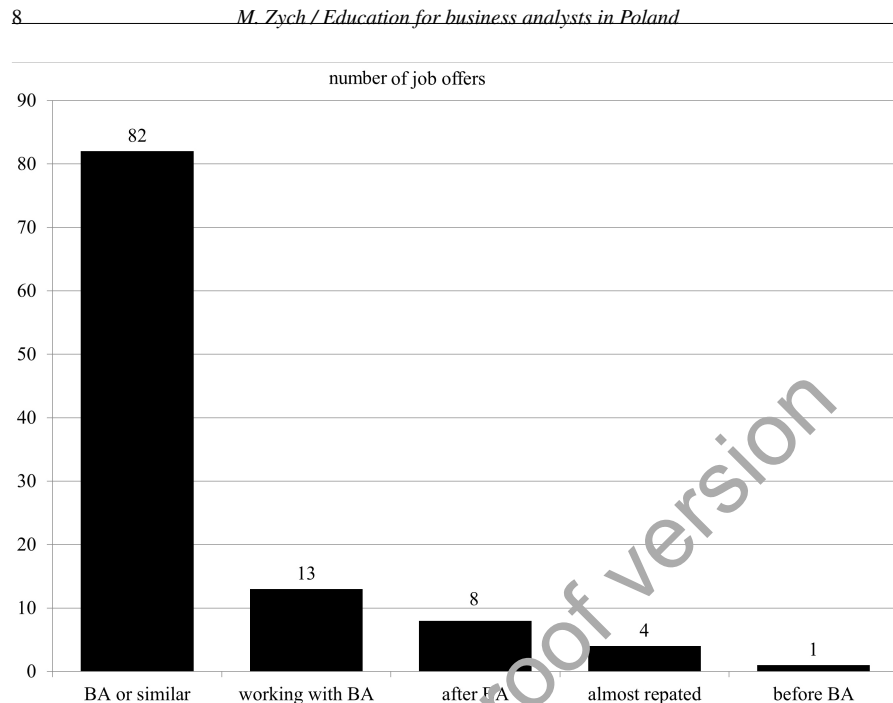


Fig. 2. Types of job positions sorted in decreasing order of BA job offers. Approximately three-quarters (82/108) of offers are labelled as 'BA or similar'.

Findings from the analysis of job offers are divided into four categories: (C1) job position name and type, (C2) type and place of employment, (C3) employer's characteristics, and (C4) skills and other requirements.

Despite the relevance ranking, LinkedIn does not filter results based on the job title only. This is fortunate for two reasons. First, it enables an initial identification of professional career paths for BAs. Second, employees working as BAs may not have the exact BA job title (e.g. IIBA, 2015, p. 2), especially in countries where recognition for BAs is still growing, such as Poland. The names of job positions from all the collected offers are shown in Fig. 1. The most popular are Junior Business Analyst (31/108), Business Analyst (21/108) and Analyst (13/108, e.g. Dispute Analyst, Financial Analyst, Continuous Improvement Analyst or TA Analyst).

Results regarding types of job positions are shown in Fig. 2. The majority of offers (82/108) are classified as 'BA or similar'. From a total of 108 open positions, 13 were labelled as 'working with BA', such as front-end/software/full-stack developers, engineers, user experience/user interface (UX/UI) designers, IT and Data Architects, and service specialists. The third group (8/108) were post-initial BA jobs ('after BA'), which means jobs that are open to those with some BA experience, such as product/project managers, senior data, process or continuous improvement analysts, and senior business specialists. There is only one offer labelled entry-level BA ('before

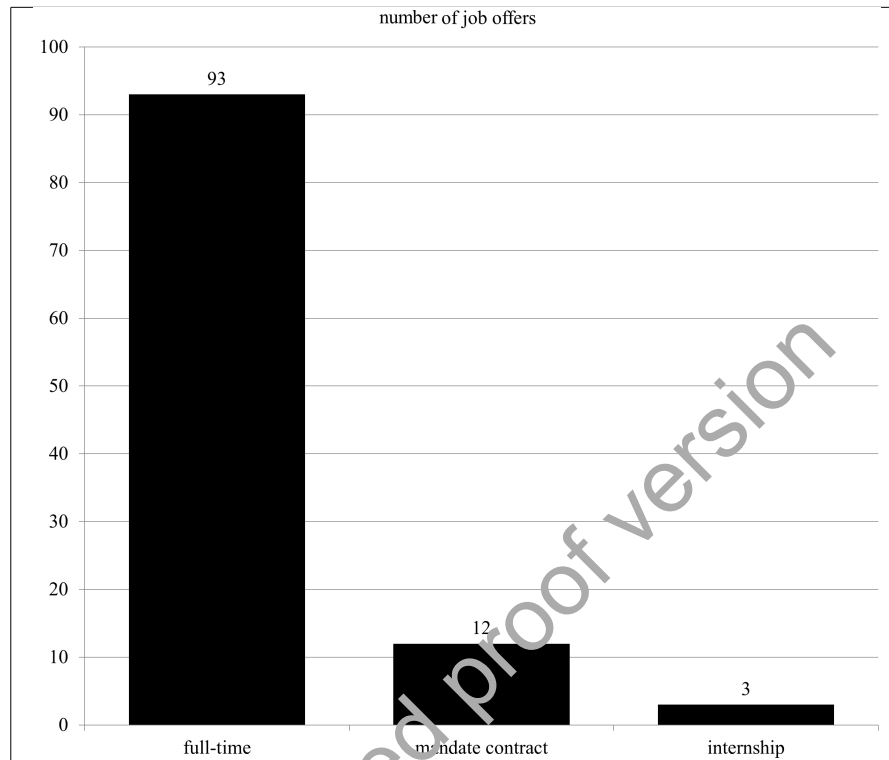


Fig. 3. Types of employment sorted in decreasing order of BA job offers. Majority (93/108) of offers are full-time jobs.

BA'), which means jobs that can be taken without any previous BA experience, such as a graduate from an IT course. Four out of 108 offers are labelled almost repeated, which means they are slightly different than originals labelled otherwise, usually in terms of place or type of employment.

Moving to the second category, type (Fig. 3) and place of employment (Fig. 4): the majority of offers are for full-time employment (93) and take place in different Polish cities such as Warszawa (50 offers), Kraków (27 offers) or Wrocław (13 offers).

There were positions defined by the characteristics of the employers: 72 discrete company names were identified, 17 of which published more than one offer, including one company with six offers, one with five offers and four companies with four published offers. The names of those companies were also collected; however, they are not disclosed in this article in order to avoid advertising them. As shown in Fig. 5, job offers came from companies of different sizes, with the biggest companies, those over 10,001 employees, publishing the most offers.

The study focuses principally on the exploration of the skills and requirements found in job offers. This is described in the following order: first, an overall view

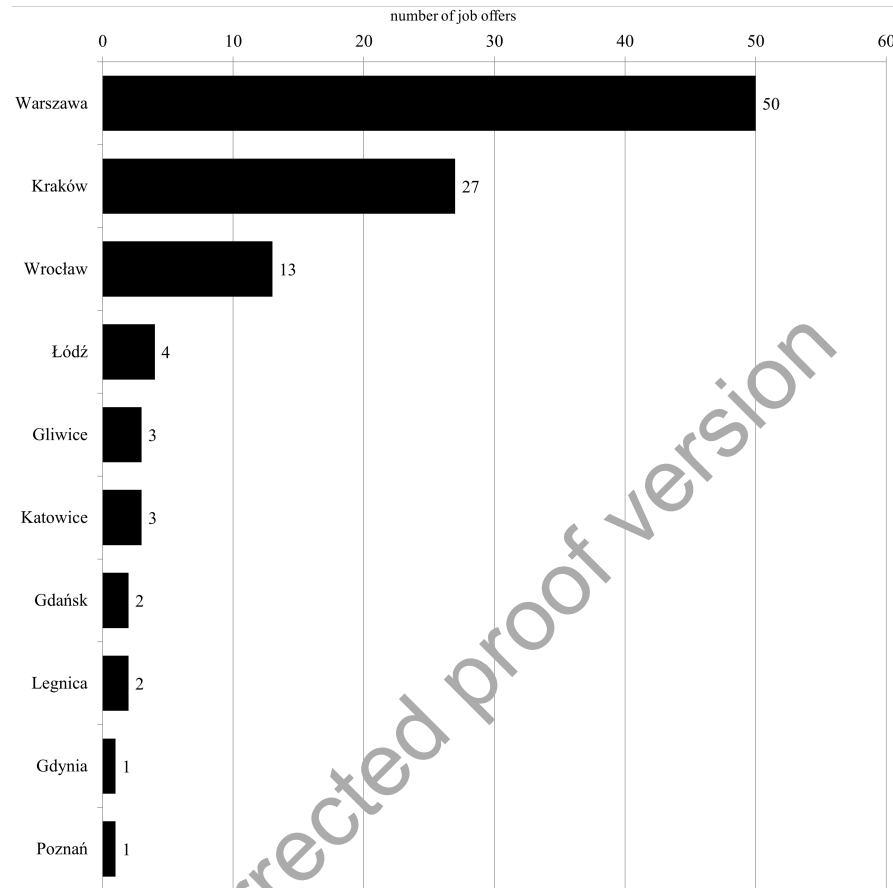


Fig. 4. Places of employment sorted in decreasing order of BA job offers. The majority of offers are for jobs located in big cities in Poland, like Warszawa (50/106), Kraków (27/106) and Wrocław (13/106). N = 106, due to incomplete records in some job offers.

is presented, second, a brief explanation of all the created code groups, and third, a mining of the details, such as the characteristics of the university degree. In addition, there are two parallel descriptions: one for the entire set of 108 collected job offers and the other for the subset of 82 job offers labelled as BA or similar.

All 196 codes for skills and other job requirements were grouped into 34 groups for the whole set and 31 groups in the BA or similar subset. There are three groups not included with regard to BA positions or similar subset: application programming interface (API), ethical hacking and information security. Table 1 shows all the code groups sorted in descending order according to the total number of job offers, both for the whole set and BA or similar subset. The order of groups is identical in both sets up to the 16th group. The first five groups are (1) language skills in terms of natural

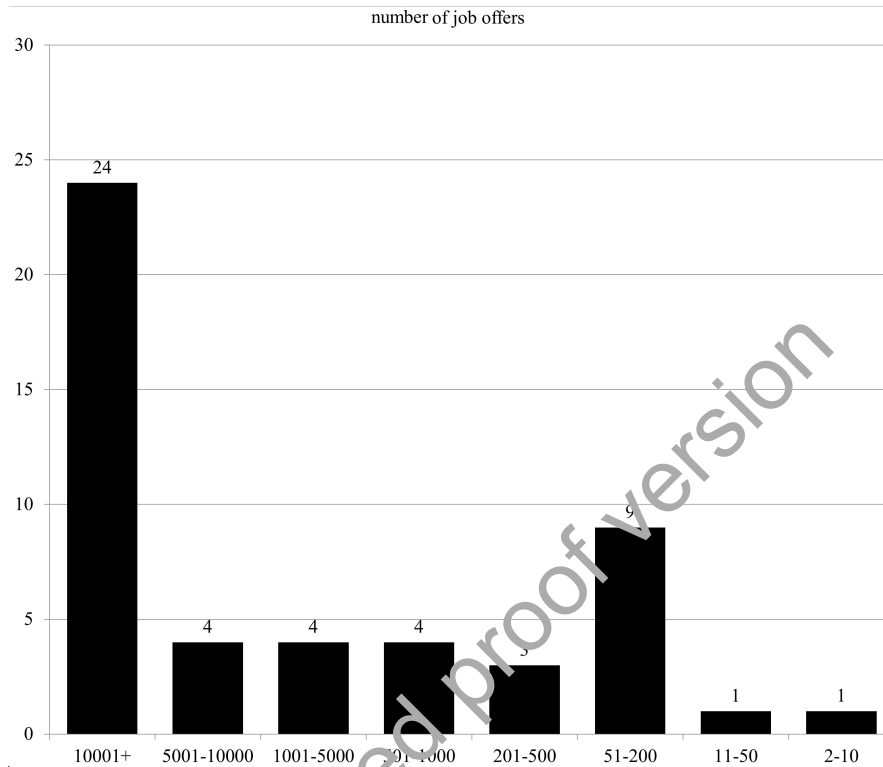


Fig. 5. Size of a company and the number of BA job offers. Approximately half of the hits with known characteristics of a company are for the biggest companies that hire over 10001 employees. $N = 50$, due to incomplete records in the set of job offers.

languages, (2) knowledge of IT tools, such as Excel, SAP or Jira, (3) communication, presentation and mediation skills, (4) personality traits, such as the ability to learn quickly, self-motivation, creativity or coping under pressure, and (5) analytical skills, risk assessment and statistics.

Besides data from Table 1, a qualitative description of each identified code group is included in order to demonstrate the whole picture of BA skills requested in Polish job offers. The following characteristics of these groups are presented in descending order of hits.

1. **Language** – knowledge of natural languages, mostly English, German, French or Italian.
2. **Tools** – knowledge of IT tools such as (1) Microsoft Office, especially Excel and Power Point, followed by Access and Share Point, occasionally Project, Word or Visio; (2) IT Service Management tools: mostly SAP, but also ServiceNow, Salesforce or SAS; (3) collaboration tools linked to agile frameworks: Jira, Confluence, sometimes Trello; (4) data analytics and business intelligence

Table 1
The code groups sorted in descending order according to the total amount of job offer

No.	Code group	BA or similar	All
1	Language	57	75
2	Tools	52	70
3	Communication, presentation, mediation skills	49	64
4	Personality traits	48	64
5	Analytical skills, risk assessment, statistics	44	54
6	Experience	40	53
7	Project, time, change, organizational management	40	53
8	Degree	40	51
9	Domain knowledge	35	44
10	Teamwork	29	42
11	Requirements engineering	27	37
12	Programming	27	37
13	Problem solving & research	24	34
14	Process framework	17	24
15	Data	17	21
16	Business understanding	15	19
17	Availability & mobility	12	16
18	BA understanding	13	13
19	IT and the latest IT	11	13
20	UX/UI	8	13
21	Client servicing skills, CRM, remote support	11	12
22	Decision making	8	12
23	Systems modelling, engineering, integration	8	10
24	Documentation skills	7	9
25	Varia	5	6
26	Process modelling	5	5
27	Database	3	3
28	Notations	3	3
29	Product development	2	3
30	API	0	2
31	Corporate values	1	2
32	Ethical hacking	0	1
33	Information security	0	1
34	SEM/SLO	1	1

tools: Power BI, Tableau, Qlik View, Qlick Sense, Microstrategy, Alteryx or Google Analytics; (5) general cloud solutions: AWS, Azure, Google Cloud Platform, Cloudera, Informatica or SailPoint; (6) Oracle – used for database management; (7) Quality Center – used for testing products; (8) Git and Git Hub; and (9) Google Sheets. There are also other tools that do not form part of BA or similar job type offers but may sometimes be required. These are Abstract app, Adobe, Ansible, AppScan, Data Power, e-commerce platforms, FogBugz, Gerrit, Highcharts, IBM Netezza, Jenkins, Linux, Liquibase, Maven, Nexus, Sonar, Spring, UX tools (Axure, Mockflow, UXPin, Wiremock), Web-pack and Wildfly.

- 3. Communication, presentation, mediation skills** – general communication skills, both written and oral, the ability to create and deliver presentations,

mediating and negotiating between the team and stakeholders, and the ability to effectively communicate between various departments of an organisation and between business and IT, for example highly communicative and able to find a common language with people of diverse backgrounds.

4. **Personality traits** – a set of skills and attitudes including the ability to learn quickly, self-motivation, demonstration of initiative, creativity, coping under pressure, responsibility, leadership skills, multitasking, open and positive attitude, ability to focus and curiosity.
5. **Analytical skills, risk assessment, statistics** – for example strong analytical skills and attention to detail, risk assessment skills, knowledge of statistics and experience using statistical packages for analysing datasets.
6. **Experience** – all those requirements that follow the scheme: a set number of years on a specific job position, for example at least one year of prior BA or QA analyst experience preferred.
7. **Project, time, change, organisational management** – mostly time and project management, but also organisational skills, followed by change management, sometimes lean management and running meetings.
8. **University degree** – this group is essential for the study, especially for hypothesis H1, since it is a direct link between required formal education and the demands of the job market. This does not necessarily mean that the identified courses are the best university courses on their own. Rather, identifying these courses allows the identification of which degrees are the most recognised and appreciated by employers. This would also aid academia who could use these listed degree courses to create benchmarks while developing new courses or redesigning existing ones. To get to the core of the issue: in 51 offers (from a total of 108), there is information on the required degree. It is a common practice to list several degrees in one offer, leaving the prospective employee the ability to choose the one that best fits. Due to this fact, the sum of hits for each code exceeds 51. For this reason, all the degrees are listed in Table 2. What is the most interesting is the following:
 - a. Information Management appears twice
 - b. Labels ‘general’ (16) and ‘similar’ (17) are common. This could mean that employers might be open to various degree courses, maybe even new ones
 - c. Employers might not value a formal degree so much, since only 51 offers among 108 specify it. Moreover, among those 51 offers, in 18 there is the label ‘or equivalent’
 - d. Those degrees mentioned come from various fields, for example Computer Science, mathematics and technology oriented, finance and economics oriented, and business and management oriented.
9. **Domain knowledge** – knowledge specific to a certain discipline or a field, this study found mostly banking, finance and insurance, but energy, health and pharmaceutical industry also appear.

Table 2

Degree programme names sorted in descending order according to the total amount of job offer

No.	Degree programme name	BA or similar	All
1	Computer Science	12	20
2	“or equivalent”	14	18
3	“similar”	14	17
4	“general”	16	16
5	Finance	11	14
6	Business	11	13
7	Economics	9	13
8	Engineering	6	10
9	Mathematics	5	6
10	Technology	4	5
11	Statistics	3	4
12	Controlling	2	3
13	Management	2	3
14	“technical”	2	3
15	Accounting	2	2
16	Data Science	1	2
17	Information Management	2	2
18	Information Security	1	2
19	Logistics	1	2
20	Marketing	2	2
21	Banking	1	1
22	Management Information Systems	1	1

10. **Teamwork** – general collaboration skills in terms of so-called soft skills, apart from knowledge of collaborative software; also, the ability to work in virtual or dispersed groups, for example team player with ability to work collaboratively and on virtual teams.
11. **Requirements engineering (RE)** – mostly prioritising requirements, general requirements that elicit skills and requirements testing (also performing User Acceptance Testing – UAT), followed by several requirements eliciting techniques: user stories, workshops, use case diagrams, event storming and product roadmap.
12. **Programming** – mostly represented by SQL and general coding understanding, occasionally: Java, HTML/CSS, Python, Angular, BEM, Ngrx, R, RxJS, TypeScript, ABAP, Angular, MDX or React.
13. **Problem-solving and research** – a willingness and ability to solve problems, perform research and ask questions, for example problem-solving skills; investigative mind-set; diagnosis of problems, and recommendation or implementation solutions; and a strong desire for research.
14. **Process framework** – agile frameworks in general, (not named) software and system development methodology, Scrum and occasionally Waterfall.
15. **Data** – all skills related to data analysis, data modelling, data management, data mining, data warehousing, data integration, dealing with large data, data

design, data governance, data mapping, data quality, data requirements, data strategy, master data objects, and dealing with missing data.

16. **Business understanding** – general business knowledge and skills, for example sound business judgement; deep understanding of the business.
17. **Availability and mobility** – business travel and availability to work flexible hours based on varying business requirements or shift work.
18. **BA understanding** – general BA knowledge and skills; knowing how to get the job done as a BA, for example BA attitude; business analysis skills.
19. **IT and the latest IT** – mostly general IT understanding and technical skills, occasionally basic understanding of Internet of Things, Information Technology Infrastructure Library (ITIL), IT Service Management (ITSM), and machine learning.
20. **UX/UI** – UX and/or UI and/or usability skills.
21. **Client servicing skills, CRM, remote support** – mostly general client servicing skills, followed by Customer Relationship Management (CRM), and one remote support mention, for example excellent client-facing skills.
22. **Decision-making** – decision-making abilities or for example service-oriented attitude balanced with ‘doing what is right for the project/business’, even if that is not a popular decision.
23. **Systems modelling, engineering, integration** – for example understanding of systems engineering concepts.
24. **Documentation skills** – ability to create documentation, for example strong documentation skills, experience with solution documentation.
25. **Varia** – set of unrelated, unreported thematic features from message queues to legal work permits.
26. **Process modelling** – for example perfect understanding of analysing and modelling business processes.
27. **Database** – general knowledge of databases.
28. **Notations** – mostly Unified Modelling Language (UML), followed by Business Process Model and Notation (BPMN), and Decision Model and Notation (DMN).
29. **Product development** – familiarity with good practices and standards in product development.
30. **API** – understanding application programming interfaces.
31. **Corporate values** – ability to apply and encourage corporate values at work.
32. **Ethical hacking** – red-teaming, penetrating systems, writing reports on findings and collaborating with owners to update systems.
33. **Information security** – understanding of information security concepts and information security certifications.
34. **SEM/SEO** – knowledge of Search Engine Marketing (SEM) and Search Engine Optimisation (SEO).

The second part of this study presents results from the comparative analysis of skills drawn from the Polish BA job description, BABOK 3, collected job offers, and

learning outcomes from a Polish LIS-related degree course (first and second cycle in Information Management), shown in Table 3. A new, modified codebook (called codebook_2) was created in order to conduct the comparison of skills between those four different sources.

Codebook_2 draws on the previous codebook created from analysing job offers. New features from BABOK 3, job description and learning outcomes that did not fit the original codes received their own codes and groups, if needed. As a result, codebook_2 includes 219 codes, grouped in 40 code groups. In Table 3, a grey field indicates that a group was identified in a source. In case of a partial hit, additional description was attached to the field in order to specify the scope of identified skills. Newly added features such as code groups and some of the codes are presented in italics.

These new features that were highlighted in the additional sources are conceptual thinking, culture preservation, due diligence, dealing with information (in terms of information seeking, information selection and familiarity with information sources) and general information management understanding, familiarity with intellectual property regulations, general knowledge of ICT and knowledge management tools.

Unsurprisingly, as shown in Table 3, there are differences among skills identified in different sources. First and foremost, sources differ in similarity to features from job offers. BABOK 3 seems the closest to the collected job offers in terms of skills required for prospective BAs. Only two sources (job offers and BABOK 3) share distinctive BA features such as, for example understanding the nature of business analysis, requirements engineering, using notations and process frameworks, and systems modelling and programming. These are also the only ones that mention willingness to undertake business travel. At first, availability and mobility may not look like ‘serious’ features, since one does not need to take either time-consuming courses or formal education to acquire those skills. Business travelling might, however, become problematic; it could impact the work-life balance, it requires decent health and an ability to travel, as well as adaptation to changing environments. With that in mind, this feature should not be neglected by prospective BAs and their future employers.

Learning outcomes from the LIS-related degree and the Polish job description score less than 50 per cent in similarity between job offers: 41.2% and 23.5% respectively. More importantly, both sources lack direct mention of distinctive BA skills covered in BABOK 3 and job offers, such as programming and requirements engineering. All four sources serve their own purposes, leading to a natural variation in depth, length and estimation date. That stated, programming seems unlikely to be overlooked, mistaken or hidden among other features.

The case of requirements engineering (RE) is probably more complex. Requirements engineering, requirements elicitation and requirements testing – these are fixed terms with concrete procedures and techniques, such as user stories, use case diagrams, event storming and product roadmap. Job offers and BABOK 3 explicitly use these terms, while learning outcomes make reference to information needs and

Table 3
Comparison of skills

No.	Code group 2	Job offers	BABOK 3	LIS-related degree	PL job description
1	Analytical skills, risk assessment, statistics	1	1	1	
2	API	1			
3	Availability & mobility	1	1		
4	BA understanding	1	1		
5	Business understanding	1	1	Only 2 nd cycle	1
6	Client servicing skills, CRM, remote support	1	1		
7	Conceptual thinking	1	1		
8	Communication, presentation, mediation skills	1	1	1	1
9	Corporate values	1	1	1	
10	Culture preservation			1	
11	Data	1	1		Data analysis
12	Database	1	1		
13	Decision making	1	1		Decision making
14	Degree	1	1		
15	Documentation skills	1	1	-	
16	Domain knowledge	1	1	1	Documentation skills
17	Due diligence	1	1		Banking and finance
18	Ethical hacking	1	1		1
19	Experience	1			
20	Information			1	
21	Information management understanding			1	
22	Information security	1		1	
23	Intellectual property			1	
24	IT and the latest IT	1	1	1	
25	Language	1		1	
26	Notations	1	1		
27	Personality traits	1	1	1	
28	Problem solving & research	1	1	1	Problem solving
29	Process framework	1	1		
30	Process modelling	1	1	1	
31	Product development	1	1	1	

Table 3, continued

No.	Code group 2	Job offers	BABOK 3	LIS-related degree	PL job description
32	Programming	1	1		
33	Project, time, change, organizational management	1	1		
34	Requirements engineering	1	1	1	Running meetings
35	SEM/SEO	1			
36	Systems modelling, engineering, integration	1	1		
37	Teamwork	1	1	1	
38	Tools	1	1		
39	UX/UI	1	1		
40	Varia	1	1		
	Total counts from code groups 2	34	28	15	9
	Similarity to job offers* (counts from code groups)	100% (34/34)	76,5% (26/34)	41,2% (14/34)	23,5% (8/34)

*Similarity to job offers is understood here as a percentage of group counts from the first codebook created based on the features solely from collected job offers.

understanding human information behaviour. These two perspectives are similar, so it is possible to expect at least some requirements engineering (RE) techniques in the LIS-related degree. However, it would require further study of the level of educational materials, since syllabuses could be too general or use different labels instead of BA names for RE procedures and techniques. Specifically, a Polish BA job description is too short to find any notion of RE here.

There are five code groups common to all four sources. These groups are (1) business understanding, (2) communication, presentation and mediation skills, (3) domain knowledge, (4) problem-solving and research, (5) project, time, change and organisational management.

All sources apart from the Polish job description emphasise general know-how: in the case of BABOK 3 and job offers, the focus is on BA understanding, whereas learning outcomes are focused on information management understanding. Moreover, these three sources stress the importance of personality traits such as creativity, curiosity, responsibility and self-learning skills.

5. Discussion

This article looks at BA skills and compares them with learning outcomes in selected LIS degrees in Poland. Findings from the analysis of job offers, BABOK 3, the job description from the Standard Classification of Occupations and learning outcomes for the first and second cycle of Information Management degrees allow verification of the following hypotheses.

H1. Employers in Poland look for their future BAs among graduates with different degrees.

Results from the analysis of job offers for BAs (see Table 2) show that employers are willing to hire graduates with a variety of degrees from Computer Science, Mathematics and Technology oriented degrees, through Finance and Economics to Business and Management. This displays a similarity to Polish higher education oriented to Business Analysis degrees, usually labelled Business Analytics, offered by faculties and departments specialising in various disciplines. Some employers are even familiar with LIS-related degrees such as Information Management, Information Security and Management Information systems. This emerging recognition looks like a good sign for LIS degrees in Poland, which are being forced to look for new research and educational paths due to institutional changes in Poland (the deregulation of librarian as a profession in 2013 and the new Law on Higher Education and Science in 2018). Therefore, up to this point, findings support hypothesis 1. Verma et al. (2019, p. 243) also report various degrees intended for future BAs. In addition, Verma et al.'s (2019, p. 249) research depicts a BA as the least technical job position in analytics jobs such as data analysts, business intelligence analysts and data scientists.

However, the study reveals something else regarding formal education required: it might not be required as much as assumed. More precisely, the study shows that

employers often do not specify a university degree in their requirements (57/108), and those who do are often open to candidates with a degree equivalent (18/51). This finding is unforeseen and does not match the expected level of education of a BA from the Standard Classification of Occupations (Ministry of Labour and Social Policy, 2018). It may be associated with the selected career stage, yet more evidence is needed to explain this result. Having said that, future studies could expand beyond requirements for an entry-level BA.

H2. LIS-related degrees provide the core competences required for a BA job in Poland.

The most common requirements in BA job offers, mainly expressed also in BABOK 3 and the job description from the Standard Classification of Occupations, are included in the learning outcomes on Information Management. These include language skills, knowledge of IT tools, communication, presentation and mediation skills, personality traits and analytical skills. A similar set of core BA skills is reported by Verma et al. (2019, p. 247), only with more specified roles of domain knowledge and structured data management, whereas Jarzębowicz and Połocka (2017, pp. 1197–1198) specify requirements engineering as the crucial skill for BAs in Poland. In general, learning outcomes from the Information Management degree share 15 codes with the BA profile created based on job offers, BABOK 3 and the job description, and its similarity to job offers is 41.2% (see Table 3). Both the BA profile and learning outcomes underline general know-how in business understanding and information management understanding.

Nevertheless, learning outcomes do not reveal certain distinctive skills of BAs found in the created BA profile, such as understanding the nature of business analysis, client servicing skills, remote support, dealing with data and databases, decision-making, documentation skills and usage of notations, information security, knowledge of process frameworks and programming. However, some of these features are taught in Information Management degrees, despite their absence in learning outcomes. This could be for two reasons: learning outcomes are created at a very general level, and there are also differences in terminology used by BAs and LIS.

Let us look at the following examples. Firstly, databases are indeed taught in several LIS subjects, for example information system designing and evaluating. Databases are considered by LIS as information systems, and their evaluation and design are part of information product development, and that product development is what learning outcomes show, not databases themselves. Secondly, dealing with data might be hidden in learning outcomes on a general level as part of dealing with information, although there are subjects in Information Management dedicated to data, such as big data, data driven-marketing and data management. Thirdly, usage of notations such as BPMN and knowledge of process frameworks such as Scrum appear on LIS with regard to process modelling and product development, for example in subjects called information process modelling, and information system designing and evaluating. In addition, RE could be understood by LIS as part of understanding information

behaviour and information needs. However, programming skills seem indeed to be lacking on the degree course.

Details such as specific tools and procedures actually required by employers are important. Santoso and Hadi Putra (2017, p. 147) state that mastering skills means not only familiarity with the principles, but also actual usage of specific tools. Besides, Jarzębowicz and Połocka (2017, p. 1197) stress the importance of the ability to choose the right tools/techniques for business problems. Gruszka (2020) also emphasises that LIS education in Poland strongly varies depending on universities, and it could go even further with educational quality depending on subjects and teachers. For example, based on learning outcomes and subject syllabus, in information system designing and evaluating, students could create simple databases in Access and websites in HTML, CSS and Wordpress with usage of basic UX procedures (e.g. card sorting). Or they could deepen and widen their skills, using SQL language and advanced joins in relation to databases, designing websites or apps in Scrum, using prototyping (e.g. Axure) and collaborative tools (e.g. Jira) with requirements elicitation techniques (e.g. user stories). The choice between these paths usually belongs to a teacher and might be visible on an educational materials level only.

To conclude, the findings support hypothesis H2, since learning outcomes in Information Management reflect the most common skills from the BA profile, and both are product development-oriented. Most of the remaining BA skills could be developed in LIS-related degrees such as Information Management, although this seems to depend mostly on the quality of classes and teachers, not on fixed learning outcomes. The lack of certain distinctive BA skills in learning outcomes could be interpreted in two ways: (1) the degree does not allow the development of such skills (e.g. programming) or (2) the degree does allow the development of such skills, but these skills are not priorities. Future studies should be done on this, including analysis of subject syllabuses and educational materials. It might be worth considering terminology unification in BA and LIS. It seems to be the focus of LIS especially to identify users' needs (e.g. Chernysheva & Shepelenko, 2018, p. 90) and requirements engineering known in BA (e.g. Jarzębowicz & Marciniak, 2017, p. 321).

6. Conclusion

Giving different names to BA roles is known in scientific literature (e.g. Ahmed et al., 2019, p. 303; Jarzębowicz & Maciniak, 2017, p. 316; Santoso & Hadi Putra, 2017, p. 146). As can be seen also from Polish job offers, BA jobs tend to be labelled differently, for example Junior Business Analyst, Business Analyst, Dispute Analyst, Financial Analyst, Continuous Improvement Analyst and TA Analyst.

Based on job offers, employers are open to candidates with degrees not only in Computer Science, Finance, Mathematics or Statistics, but also in for example Business, Controlling, Management, Data Science, Information Management, Information Security or Marketing, which supports hypothesis H1. The analysed standards allow

this variety of degrees too. Furthermore, using labels such as ‘general’, ‘similar’ or ‘equivalent’ might indicate that employers are searching for either the right degree course or similar sets of skills that can be acquired outside formal education.

The most common requirements posted in job offers, which are also visible in learning outcomes for Information Management, are language skills, knowledge of IT tools (e.g. Excel, SAP and Jira), communication, presentation and mediation skills, personality traits (e.g. ability to learn quickly, self-motivation, creativity and coping under pressure) and analytical skills. Similar basic BA skills are reported in the literature (e.g. Verma et al., 2019, p. 247).

Differences regarding skills can be found between sources. Obviously, all four sources of skills vary in their level of detail and length, with learning outcomes and Polish job descriptions being the most general. With that in mind, distinctive BA features such as understanding the nature of business analysis, requirements engineering, using notations and process frameworks, systems modelling and programming can be found only in job offers and BABOK 3. Most of these remaining BA skills might be incorporated in existing LIS degrees; however, it seems to depend mostly on the quality of classes and teachers, not on fixed learning outcomes.

A basic recommendation for universities in Poland aiming at addressing the mismatch between job market demands and LIS-related degree courses is that a consultation should be conducted with potential employers. Based on this cooperation and taking into account the framework of fixed outcome-based requirements, changes in the courses can be carried out. First, breadth courses could be used to cover skills from domain knowledge, AI, data science or programming (e.g. Rouvrais & Gaultier Le Bris, 2018, pp. 327–329). Second, it might also be worth teaching tools, procedures and techniques actually required by employers, such as Excel, Jira, and user stories, for example. Since learning outcomes and syllabuses are difficult to change, it might be better to incorporate these tools into the educational materials. Third, as can be seen in the collected job offers, employers are usually large companies operating in cities, which agrees with findings from the literature (e.g. Chernysheva & Schepelenko, 2013, p. 87). It might be worth targeting such companies in order to encourage cooperation. Four, if skills such as working in process frameworks and dealing with databases were actually taught on the degree courses, it could be beneficial to add them to learning outcomes, leaving specifics to educational materials. Finally, knowledge of several languages is important. If degree courses are to provide targeted courses for developing BA skills, then appropriate business terminology should be adopted for more effective communication with companies and prospective students.

7. Limitations

The results of the analysis of the job offers visible on LinkedIn between December 2019 and January 2020 demonstrate employers’ requirements for prospective BAs

in Poland. Future studies might include in-depth interviews with Polish BAs and their employers, as well as longitudinal analysis of job offers and analysis of subject syllabuses and educational materials.

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